MOHAMMED AKMAD THARIK A

Electronics and Communication Engineering

mohammed-akmad-tharik-a-7143532a2
mohammedakmadtharika@gmail.com
+91 9751208786
github.com/Akmadtharik

Education

Bachelor of Engineering – Electronics and Communication Engineering 2023–2027
Knowledge Institute of Technology, Salem CGPA: 7.795/10

Higher Secondary (12th) 2023
Equitas Gurukul Matriculation Higher Secondary School, Salem 82.16%

Technical Skills

Electronics: Digital Electronics, Electronic Devices, Basic Circuit Troubleshooting
Embedded Systems: Microcontrollers, Sensor Interfacing, Embedded Systems, Hardware Debugging, Circuit Testing, Prototype Development
Communication Interfaces: UART, SPI, I2C, GPIO, ADC, PWM
Programming: C, Embedded C, Java
Development Boards: ESP32, ESP8266, Raspberry Pi 4, Arduino Uno/Nano, RP2040 Pico W
Tools: Arduino IDE, KiCad, EasyEDA, LTspice, VS Code, Git Hub

Core Competencies

Circuit Debugging, Hardware Testing, Sensor Interfacing, Embedded System Development, Troubleshooting, Component Identification, Prototype Development, Technical Documentation, Team Collaboration, Communication, Problem Solving

Internships

Electronics Technician & Project Intern Mar 2025 – Oct 2025
Knowledge Electronics, Salem

- Assisted with sensor and embedded peripheral integration using UART, SPI, and I2C.
- Supported circuit testing, component selection, hardware debugging, and troubleshooting.
- Assisted with project development and hardware validation activities.

IoT Development Intern Dec 2024 – Jan 2025
Airavad Web Solutions, Salem

- Developed and tested an RF-based Automatic Vehicle Speed Control and Monitoring System.
- Worked on transmitter and receiver modules for wireless speed-limit communication.
- Participated in hardware-software integration, testing, and troubleshooting.

Projects

Air Travel Assistance Band – Wearable Embedded System

- Designed an ESP32-S3 based wearable system for airport navigation and passenger assistance.
- Integrated sensors and communication peripherals for real-time system functionality.
- Worked on hardware integration, firmware flow, testing, and system improvement.

Desk Buddy Companion – Embedded Desktop Assistant

- Designed and developed an interactive desktop companion using microcontroller-based hardware.
- Integrated display, motion, audio, and environmental sensing features.
- Improved system functionality through testing, debugging, and iterative development.

RF-Based Automatic Vehicle Speed Control System

- Developed a wireless RF transmitter-receiver system for communicating roadside speed limits to vehicles.
- Designed and tested transmitter and receiver modules for reliable wireless communication.
- Integrated and tested the system for automatic speed-control functionality.

Selected Mini Projects

8×8 Matrix Wi-Fi Billboard, DIY Bluetooth Audio System, Autonomous Fire-Fighting Node, RF Transceiver, Retro Gaming Node

Certifications

- Cloud Computing – NPTEL
- Embedded Systems Design – Industry Training Program
- Digital Design Verification with SystemVerilog – HCL GUVI
- Full Stack Development – VEI Technologies
- Machine Learning Basics – Imagecon Academy
- IoT Development – Airavad Web Solutions

Achievements

- Secured 2nd place in Tamil Nadu in an e-bike manufacturing and racing event.
- Selected as a finalist team in an engineering design challenge with an embedded-system project.
- Participated in technical and engineering project activities involving embedded systems and electronics.

Languages

Tamil — English — Arabic (Basic Reading) — Japanese (JLPT N5 – Pursuing)